COURSE CODE	COURSE TITLE	L	T	P	TW & SL	TH	C
UCS3511	NETWORKS LABORATORY	0	0	45	0	45	1.5

OBJECTIVES

- To find the network statistics and understand the speed and traffic on the network.
- To guide students in simulating the operations of the data link layer, IP layer, and transport layer.
- To simulate routing, congestion control algorithms and wireless topologies.

SUGGESTED EXPERIMENTS

1. Apply the following networking tools to analyse the status, performance and record the observations. a. netstat, ifconfig, iwconfig, ping, route, nmap, tcpdump, ping, dig, traceroute, mtr.
2. Develop client server applications using TCP and UDP sockets. Modify server to handle multiple clients concurrently.
3. Simulate frame transmission at the datalink layer using SimPy, incorporating error detection mechanisms like checksum or CRC.
4. Simulate the Automatic Repeat Request (ARQ) protocol in SimPy with the receiver acknowledging the receipt of a correct frame and request retransmission for corrupted or lost frames.
5. Simulate the behavior of TCP congestion control algorithms such as Reno, Cubic, and BBR in a network using SimPy.
6. Simulate a UDP packet transmission system in a network using SimPy. Model packet generation, transmission, packet loss, and random delays.
7. Simulate a network with multiple nodes and define routing paths based on IP addresses using SimPy.
8. Simulate the ARP (Address Resolution Protocol) in a network using SimPy, where nodes map IP addresses to MAC addresses.
9. Simulate a wireless network using SimPy with multiple nodes communicating with each other. Assign Wi-Fi, Bluetooth, and Zigbee technologies to different nodes.
10. Simulate the Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) protocol used in wireless networks using SimPy.
11. Simulate TCP flow control in a network using SimPy, where the sender adjusts its transmission rate based on the receiver's advertised window size
12. Simulate common topologies, generate traffic, in Mininet and trace packet-level behaviour using Wireshark
13. Develop a simulation in Mininet to implement a simple SDN controller.
14. Mini projects like,
    - Develop a custom ARQ-based reliable data transfer mechanism in SimPy with enhancements like timeout optimization or adaptive window sizing.
    - Simulate ARP resolution and create a mechanism to detect conflicting MAC address mappings (ARP spoofing detection) using SimPy.
    - Demonstrate adaptive routing and redundancy techniques in fault-tolerant network design.
    - Develop a simple packet sniffer using tcpdump or scapy, and build a dashboard that analyzes captured traffic by protocol, source/destination, and packet type.

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- Build a tool using ping, traceroute, and mtr to measure latency and jitter between nodes. Use matplotlib or Dash to visualize performance trends over time.
- Simulate a DHCP server-client interaction in SimPy where clients request and lease IP addresses from a pool, including lease expiration and renewal.
- Simulate multiple IoT nodes using lightweight MQTT protocol over UDP and evaluate packet loss, delay, and delivery guarantees.

TOTAL PERIODS	**45**

Software: C / Python / Equivalent Compiler.

COURSE OUTCOMES
On successful completion of this course, students will be able to
CO1 Analyze network performance using tools and develop multi-client client-server applications using TCP and UDP socket programming. (K4)
CO2 Simulate core networking protocols and communication mechanisms (wired and wireless) using SimPy to model data link, transport, and routing behaviour and analyse the performance. (K4)
CO3 Design and execute a mini project to investigate complex networking scenarios using SimPy, incorporating protocol modeling, performance evaluation, and creative problem-solving. (K6)

CO-PO-PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	2	3	1							3	2
CO2	3	3	2	3	1							3	2
CO3	3	3	2	3	1	2		3	2		3	3	3

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